

Electromagnetism & Electrodynamics HW5

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Technion

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1 A 4 potential

1.1 finding the fields

We'll find the fields tensor instead and derive the fields from it. as we know, the fields tensor is:

$$F^{\mu\nu} = \partial^\mu A^\nu - \partial^\nu A^\mu \quad (1)$$

so

$$E_i = F_{0i} = \partial_0 A_i - \partial_i A_0 \quad (2)$$

$$= \partial_0 \eta_{i\alpha} A^\alpha - \partial_i \eta_{0\beta} A^\beta \quad (3)$$

$$= -\partial_0 A^i - \partial_i A^0 \quad (4)$$

$$= -\frac{\partial}{\partial x^0} \left(x^i e^{-\alpha((x^0)^2+(x^1)^2+(x^2)^2+(x^3)^2)} \right) - \frac{\partial}{\partial x^i} \left(x^0 e^{-\alpha((x^0)^2+(x^1)^2+(x^2)^2+(x^3)^2)} \right) \quad (5)$$

$$= 2\alpha x^0 x^i e^{-\alpha((x^0)^2+(x^1)^2+(x^2)^2+(x^3)^2)} + 2\alpha x^0 x^i e^{-\alpha((x^0)^2+(x^1)^2+(x^2)^2+(x^3)^2)} \quad (6)$$

$$= 4\alpha x^0 x^i e^{-\alpha((x^0)^2+(x^1)^2+(x^2)^2+(x^3)^2)} \quad (7)$$

and

$$B_1 = F_{32} = \partial_3 A_2 - \partial_2 A_3 \quad (8)$$

$$= \partial_3 \eta_{22} A^2 - \partial_2 \eta_{33} A^3 \quad (9)$$

$$= \partial_2 A^3 - \partial_3 A^2 \quad (10)$$

$$= \partial_2 \left(x^3 e^{-\alpha((x^0)^2+(x^1)^2+(x^2)^2+(x^3)^2)} \right) - \partial_3 \left(x^2 e^{-\alpha((x^0)^2+(x^1)^2+(x^2)^2+(x^3)^2)} \right) \quad (11)$$

$$= -2\alpha x^3 x^2 e^{-\alpha((x^0)^2+(x^1)^2+(x^2)^2+(x^3)^2)} + 2\alpha x^2 x^3 e^{-\alpha((x^0)^2+(x^1)^2+(x^2)^2+(x^3)^2)} \quad (12)$$

$$= 0 \quad (13)$$

from symmetry to choice of index,

$$B_2 = B_3 = 0 \quad (14)$$

so

$$\mathbf{E} = 4\alpha x^0 e^{-\alpha((x^0)^2+(x^1)^2+(x^2)^2+(x^3)^2)} \mathbf{x} \quad \mathbf{B} = 0 \quad (15)$$

1.2 4-current

Ee saw a relation between the 4-current and the fields tensor

$$\frac{4\pi}{c} \mathbf{J}^\nu = \partial_\mu F^{\mu\nu} \quad (16)$$

so

$$J^0 = \frac{c}{4\pi} \partial_\mu F^{\mu 0} \quad (17)$$

$$= \frac{c}{4\pi} \partial_i F^{i0} \quad (18)$$

$$= \frac{c}{4\pi} \partial_i \left(4\alpha x^0 x^i e^{-\alpha((x^0)^2 + (x^1)^2 + (x^2)^2 + (x^3)^2)} \right) \quad (19)$$

$$= \frac{c}{\pi} \alpha x^0 \left(3e^{-\alpha((x^0)^2 + (x^1)^2 + (x^2)^2 + (x^3)^2)} - 2\alpha x^i x_i e^{-\alpha((x^0)^2 + (x^1)^2 + (x^2)^2 + (x^3)^2)} \right) \quad (20)$$

$$= \frac{\alpha c}{\pi} x^0 (3 - 2\alpha x^i x_i) e^{-\alpha((x^0)^2 + (x^1)^2 + (x^2)^2 + (x^3)^2)} \quad (21)$$

and

$$J^i = \frac{c}{4\pi} \partial_\mu F^{\mu i} \quad (22)$$

$$= \frac{c}{4\pi} \partial_0 F^{0i} \quad (23)$$

$$= -\frac{c}{4\pi} \partial_0 \left(4\alpha x^0 x^i e^{-\alpha((x^0)^2 + (x^1)^2 + (x^2)^2 + (x^3)^2)} \right) \quad (24)$$

$$= -\frac{\alpha c}{\pi} \left(x^i e^{-\alpha((x^0)^2 + (x^1)^2 + (x^2)^2 + (x^3)^2)} - 2\alpha x^0 x_0 x^i e^{-\alpha((x^0)^2 + (x^1)^2 + (x^2)^2 + (x^3)^2)} \right) \quad (25)$$

$$= -\frac{\alpha c}{\pi} x^i (1 - 2\alpha x^0 x_0) e^{-\alpha((x^0)^2 + (x^1)^2 + (x^2)^2 + (x^3)^2)} \quad (26)$$

therefore

$$\mathbf{J}^\mu = \frac{\alpha c}{\pi} \begin{pmatrix} 3 - 2\alpha x^i x_i \\ 2\alpha x^0 x_0 - 1 \\ 2\alpha x^0 x_0 - 1 \\ 2\alpha x^0 x_0 - 1 \end{pmatrix} \mathbf{x} e^{-\alpha((x^0)^2 + (x^1)^2 + (x^2)^2 + (x^3)^2)} \quad (27)$$

We will show conservation of charge exists by showing the continuity equation works. explicitly the 4-current is

$$\mathbf{J}^\mu = \frac{\alpha c}{\pi} \begin{pmatrix} 3x^0 - 2\alpha x^0 x^i x_i \\ 2\alpha x^0 x_0 x^1 - x^1 \\ 2\alpha x^0 x_0 x^2 - x^2 \\ 2\alpha x^0 x_0 x^3 - x^3 \end{pmatrix} \mathbf{x} e^{-\alpha((x^0)^2 + (x^1)^2 + (x^2)^2 + (x^3)^2)} \quad (28)$$

so

$$\partial_\mu \mathbf{J}^\mu = \frac{\alpha c}{\pi} \frac{\partial}{\partial x^\mu} \begin{pmatrix} 3x^0 - 2\alpha x^0 x^i x_i \\ 2\alpha x^0 x_0 x^1 - x^1 \\ 2\alpha x^0 x_0 x^2 - x^2 \\ 2\alpha x^0 x_0 x^3 - x^3 \end{pmatrix} \mathbf{x} e^{-\alpha((x^0)^2 + (x^1)^2 + (x^2)^2 + (x^3)^2)} \quad (29)$$

$$= \frac{\alpha c}{\pi} \begin{pmatrix} \partial_0 & \partial_1 & \partial_2 & \partial_3 \end{pmatrix} \begin{pmatrix} 3x^0 - 2\alpha x^0 x^i x_i \\ 2\alpha x^0 x_0 x^1 - x^1 \\ 2\alpha x^0 x_0 x^2 - x^2 \\ 2\alpha x^0 x_0 x^3 - x^3 \end{pmatrix} \mathbf{x} e^{-\alpha((x^0)^2 + (x^1)^2 + (x^2)^2 + (x^3)^2)} \quad (30)$$

$$= \frac{\alpha c}{\pi} (3 - 2\alpha x^i x_i) (1 - 2\alpha (x^0)^2) e^{-\alpha((x^0)^2 + (x^1)^2 + (x^2)^2 + (x^3)^2)} \quad (31)$$

$$+ \frac{\alpha c}{\pi} (2\alpha x^0 x_0 - 1) (1 - 2\alpha (x^1)^2) e^{-\alpha((x^0)^2 + (x^1)^2 + (x^2)^2 + (x^3)^2)} \quad (32)$$

$$+ \frac{\alpha c}{\pi} (2\alpha x^0 x_0 - 1) (1 - 2\alpha (x^2)^2) e^{-\alpha((x^0)^2 + (x^1)^2 + (x^2)^2 + (x^3)^2)} \quad (33)$$

$$+ \frac{\alpha c}{\pi} (2\alpha x^0 x_0 - 1) (1 - 2\alpha (x^3)^2) e^{-\alpha((x^0)^2 + (x^1)^2 + (x^2)^2 + (x^3)^2)} \quad (34)$$

$$= \frac{\alpha c}{\pi} (2\alpha x^i x_i) (1 - 2\alpha (x^0)^2) e^{-\alpha((x^0)^2 + (x^1)^2 + (x^2)^2 + (x^3)^2)} \quad (35)$$

$$+ \frac{\alpha c}{\pi} (2\alpha x^0 x_0 - 1) (-2\alpha (x^1)^2) e^{-\alpha((x^0)^2 + (x^1)^2 + (x^2)^2 + (x^3)^2)} \quad (36)$$

$$+ \frac{\alpha c}{\pi} (2\alpha x^0 x_0 - 1) (-2\alpha (x^2)^2) e^{-\alpha((x^0)^2 + (x^1)^2 + (x^2)^2 + (x^3)^2)} \quad (37)$$

$$+ \frac{\alpha c}{\pi} (2\alpha x^0 x_0 - 1) (-2\alpha (x^3)^2) e^{-\alpha((x^0)^2 + (x^1)^2 + (x^2)^2 + (x^3)^2)} \quad (38)$$

$$= 0 \quad (39)$$

1.3 Temporal Gauge

We want to find a gauge transformation Λ such that $A^0 + \partial^0 \Lambda = 0$. Explicitly,

$$x^0 e^{-\alpha((x^0)^2 + (x^1)^2 + (x^2)^2 + (x^3)^2)} + \frac{\partial \Lambda}{\partial x^0} = 0 \quad (40)$$

therefore

$$\Lambda = - \int dx^0 x^0 e^{-\alpha((x^0)^2 + (x^1)^2 + (x^2)^2 + (x^3)^2)} = \frac{1}{2\alpha} e^{-\alpha((x^0)^2 + (x^1)^2 + (x^2)^2 + (x^3)^2)} + C \quad (41)$$

Λ has gauge freedom for constant values so we'll choose $C = 0$ and get

$$\Lambda = \frac{1}{2\alpha} e^{-\alpha((x^0)^2 + (x^1)^2 + (x^2)^2 + (x^3)^2)} \quad (42)$$

Next we'll find all the derivatives of Λ :

$$\partial_\mu \Lambda = -x_\mu e^{-\alpha((x^0)^2 + (x^1)^2 + (x^2)^2 + (x^3)^2)} \quad (43)$$

raising the index

$$\partial^\mu \Lambda = \begin{pmatrix} -x^0 \\ x^1 \\ x^2 \\ x^3 \end{pmatrix} e^{-\alpha((x^0)^2+(x^1)^2+(x^2)^2+(x^3)^2)} \quad (44)$$

so the potential is

$$A'^\mu = A^\mu + \partial^\mu \Lambda = x^\mu e^{-\alpha((x^0)^2+(x^1)^2+(x^2)^2+(x^3)^2)} \quad (45)$$

$$+ \begin{pmatrix} -x^0 \\ x^1 \\ x^2 \\ x^3 \end{pmatrix} e^{-\alpha((x^0)^2+(x^1)^2+(x^2)^2+(x^3)^2)} \quad (46)$$

$$= 2x^i e^{-\alpha((x^0)^2+(x^1)^2+(x^2)^2+(x^3)^2)} \quad (47)$$

1.4 Switching potential

noticing that the new potential can be written as

$$A^\mu = x^\mu e^{-\alpha \eta_{\alpha\beta} x^\alpha x^\beta} \quad (48)$$

and that this looks similar to the form of the gauge transformation we used in the previous subsection. We'll try to fit a gauge transformation of a similar form and see what we get. We'll try

$$\Lambda = -\frac{1}{2\alpha} e^{-\alpha \eta_{\alpha\beta} x^\alpha x^\beta} \quad (49)$$

This gives us the following derivatives:

$$\partial_\mu \Lambda = -\frac{1}{2\alpha} \eta_{\mu\gamma} \partial^\gamma \Lambda \quad (50)$$

$$= \eta_{\mu\gamma} x^\gamma e^{-\alpha \eta_{\alpha\beta} x^\alpha x^\beta} \quad (51)$$

$$= x_\gamma e^{-\alpha \eta_{\alpha\beta} x^\alpha x^\beta} = A_\gamma \quad (52)$$

We found that we can define a gauge transformation that will just cancel out the potential giving us

$$A'_\mu = A_\mu + \partial_\mu \Lambda = 0 \quad (53)$$

And since the potential is zero, the fields tensor will also be 0, and since it's invariant for gauge transformations, we find that

$$\mathbf{E} = 0 \qquad \mathbf{B} = 0 \quad (54)$$

2 Show Things

2.1 Deriving that equation

$$\partial_\mu T^\mu_\nu = \partial_\mu \left(-\frac{1}{4\pi c} F^{\mu\rho} F_{\nu\rho} + \frac{1}{16\pi c} \delta^\mu_\nu F^{\rho\sigma} F_{\rho\sigma} \right) \quad (55)$$

$$= \frac{1}{4\pi c} \left(\frac{1}{4} \partial_\mu (\delta^\mu_\nu F^{\rho\sigma} F_{\rho\sigma}) - \partial_\mu (F^{\mu\rho} F_{\nu\rho}) \right) \quad (56)$$

$$= \frac{1}{4\pi c} \left(\frac{1}{4} (\partial_\nu F^{\rho\sigma} F_{\rho\sigma} + F^{\rho\sigma} \partial_\nu F_{\rho\sigma}) - (\partial_\mu F^{\mu\rho} F_{\nu\rho} + F^{\mu\rho} \partial_\mu F_{\nu\rho}) \right) \quad (57)$$

$$= \frac{1}{4\pi c} \left(\frac{1}{2} F^{\rho\sigma} \partial_\nu F_{\rho\sigma} + \partial_\mu F^{\rho\mu} F_{\nu\rho} + F^{\rho\mu} \partial_\mu F_{\nu\rho} \right) \quad (58)$$

$$= \frac{1}{4\pi c} \left(\frac{1}{2} F^{\rho\sigma} \partial_\nu F_{\rho\sigma} - \frac{4\pi}{c} J^\rho F_{\nu\rho} + F^{\rho\mu} \partial_\mu F_{\nu\rho} \right) \quad (59)$$

$$= -\frac{1}{c^2} F_{\nu\rho} J^\rho + \frac{1}{8\pi c} F^{\rho\sigma} \partial_\nu F_{\rho\sigma} + \frac{1}{4\pi c} F^{\rho\mu} \partial_\mu F_{\nu\rho} \quad (60)$$

$$= -\frac{1}{c^2} F_{\nu\rho} J^\rho + \frac{1}{8\pi c} (F^{\rho\sigma} \partial_\nu F_{\rho\sigma} + F^{\rho\mu} \partial_\mu F_{\nu\rho} + F^{\rho\mu} \partial_\mu F_{\nu\rho}) \quad (61)$$

$$= -\frac{1}{c^2} F_{\nu\rho} J^\rho + \frac{1}{8\pi c} F^{\rho\mu} (\partial_\nu F_{\rho\mu} + \partial_\mu F_{\nu\rho} + \partial_\mu F_{\nu\rho}) \quad (62)$$

$$= -\frac{1}{c^2} F_{\nu\rho} J^\rho \quad (63)$$

2.2 conservation rules

2.2.1 for $\nu = 0$

The equation we start with is

$$\partial_\mu T^\mu_0 = \frac{1}{c^2} F_{0\mu} J^\mu \quad (64)$$

Expanding we see

$$\partial_\mu T^\mu_0 = \partial_0 T^0_0 + \partial_i T^i_0 \quad (65)$$

$$= -\frac{1}{c^2} F_{00} J^0 + \partial_i T^i_0 \quad (66)$$

$$= \partial_i T^i_0 = -\frac{1}{c^2} F_{0i} J^i \quad (67)$$

$$(68)$$

therefore

$$\frac{\partial E}{\partial t} + \partial_i T^i_0 = -\frac{1}{c^2} F_{0i} J^i \quad (69)$$

recalling the definition of the 4-current, fields tensor, and momentum-energy tensor we can rewrite that as

$$\frac{\partial E}{\partial t} + \nabla \cdot \mathbf{S} = -\mathbf{E} \cdot \mathbf{j} \quad (70)$$

as required.

2.2.2 for $\nu = i$

The equation we start with is

$$\partial_\mu T^\mu{}_i = \frac{1}{c^2} F_{i\mu} J^\mu \quad (71)$$

expanding we get

$$\partial_\mu T^\mu{}_i = \partial_0 T^0{}_i + \partial_j T^j{}_i \quad (72)$$

$$= \frac{1}{c} \frac{\partial \mathbf{S}}{\partial t} + \nabla \cdot \overleftrightarrow{\sigma} \quad (73)$$

$$= \partial_t \mathbf{p}_{em} + \nabla \cdot \overleftrightarrow{\sigma} \quad (74)$$

and on the other hand:

$$\frac{1}{c^2} F_{i\mu} J^\mu = \frac{1}{c^2} F_{i0} J^0 + \frac{1}{c^2} F_{ij} J^j \quad (75)$$

$$= -\frac{1}{c^2} E_i c \rho + \frac{1}{c^2} \epsilon^k{}_{ji} B_k J^j \quad (76)$$

$$= -\frac{1}{c^2} E_i c \rho + (B \times \mathbf{J})_i \quad (77)$$

$$= -\frac{1}{c} \left(E_i \rho + \frac{1}{c} \mathbf{J} \times \mathbf{B} \right) \quad (78)$$

combining

$$\partial_t \mathbf{p}_{em} + \nabla \cdot \overleftrightarrow{\sigma} = -\frac{1}{c} \left(E_i \rho + \frac{1}{c} \mathbf{J} \times \mathbf{B} \right) \quad (79)$$

$$\partial_t \mathbf{p}_{em} + \frac{1}{c} \left(E_i \rho + \frac{1}{c} \mathbf{J} \times \mathbf{B} \right) = \nabla \cdot \overleftrightarrow{\sigma} \quad (80)$$

$$\partial_t \mathbf{p}_{em} + \partial_t \mathbf{p}_{mech} = \nabla \cdot \overleftrightarrow{\sigma} \quad (81)$$

3 Moving Plane

The electric field is

$$\mathbf{E} = \begin{cases} 2\pi\sigma\hat{\mathbf{x}} & x > vt \\ -2\pi\sigma\hat{\mathbf{x}} & x < vt \end{cases} \quad (82)$$

but looking at the definition of the stress tensor

$$\sigma_{ij} = \frac{1}{4\pi} \left(E_i E_j + B_i B_j - \frac{1}{2} \delta_{ij} (E^2 + B^2) \right) \quad (83)$$

we can immediately see that all the off-diagonal elements are 0, and the tensor is from a direction calculation is

$$\sigma_{ij} = \frac{1}{8\pi} \begin{pmatrix} (2\pi\sigma)^2 - B^2 & & \\ & -((2\pi\sigma)^2 + B^2) & \\ & & B^2 - (2\pi\sigma)^2 \end{pmatrix} \quad (84)$$

which is constant in space.

3.1 Stored Momentum

The EM momentum density is defined as

$$\mathbf{p}_{EM} = \frac{1}{c^2} \mathbf{S} = \frac{1}{4\pi c} \mathbf{E} \times \mathbf{B} \quad (85)$$

so

$$\mathbf{p}_{EM} = \begin{cases} \mathbf{p}_{EM,>} = -\frac{\sigma B}{2c} \hat{\mathbf{y}} & x > vt \\ \mathbf{p}_{EM,<} = \frac{\sigma B}{2c} \hat{\mathbf{y}} & x < vt \end{cases} \quad (86)$$

now that we have the density on each side, to get the total we multiply by the volume on each side

$$\mathbf{p}_{EM}^{Box} = \mathbf{p}_{EM,>} \cdot A(h + vt) + \mathbf{p}_{EM,<} \cdot A(h - vt) = \frac{AB\sigma vt}{c} \hat{\mathbf{y}} \quad (87)$$

3.2 Force per area

First noting that since $\overleftrightarrow{\sigma}$ is constant, the contribution from it's sum over all the faces of the box will be zero, means we can simplify to

$$\frac{d\mathbf{p}_{Mech}^{Box}}{dt} = -\frac{d\mathbf{p}_{EM}^{Box}}{dt} \quad (88)$$

inserting our result

$$\frac{d\mathbf{p}_{Mech}^{Box}}{dt} = -\frac{AB\sigma v}{c} \hat{\mathbf{y}} \quad (89)$$

so

$$\frac{F}{A} = -\frac{B\sigma v}{c} \hat{\mathbf{y}} \quad (90)$$